Day 4: Firewall BasicsTheory

Task 1 : Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Procedure

Step 1: Open Windows Defender Firewall

1. Open the Windows Start menu.
2. Search for **Windows Defender Firewall with Advanced Security**.
3. Open the application.

Step 2: Open Outbound Rules

1. Select **Outbound Rules** from the left-side panel.
2. Click **New Rule...** on the right-side panel.

Outbound rules control network connections initiated by applications on the computer.

Step 3: Select the Rule Type

1. Select **Program**.
2. Click **Next**.
3. Choose **This program path**.
4. Browse to or enter the actual path of the Spotify desktop executable, Spotify.exe.
5. Click **Next**.

Step 4: Block the Connection

1. Select **Block the connection**.
2. Click **Next**.

This action instructs Windows Firewall to block outgoing connections matching the rule.

Step 5: Select Network Profiles

Select the following profiles:

- Domain
- Private

- Public

Click **Next** to continue.

Step 6: Name the Rule

1. Enter the rule name: Block Spotify Outbound.
2. Optionally enter a description: Blocks outgoing network traffic from Spotify.
3. Click **Finish**.

Step 7: Verify the Rule

1. Find Block Spotify Outbound in the Outbound Rules list.
2. Confirm that the rule is enabled.
3. Open Spotify and try playing an online song.

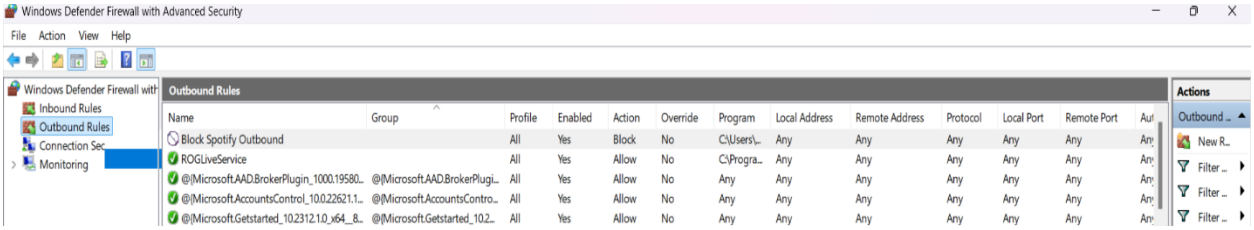
Practical Observation

When the outbound rule was enabled, Spotify opened, but online songs did not play. After the rule was disabled, song playback worked again.

This indicates that the firewall rule was responsible for blocking Spotify's online connectivity during the test.

Result

An outbound firewall rule was successfully created for the Spotify desktop application using Windows Defender Firewall.



Task 2 : Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

To configure Windows Defender Firewall to allow incoming TCP connections for Google Chrome on port 443.

Procedure

Step 1: Open Windows Defender Firewall

Open the Start menu, search for **Windows Defender Firewall with Advanced Security**, and launch it.

Step 2: Create a New Inbound Rule

Select **Inbound Rules** from the left panel and click **New Rule...**.

Step 3: Select the Program

1. Choose **Program** and click Next.
2. Select **This program path**.
3. Browse to the actual chrome.exe file.
4. Click Next.

Step 4: Allow the Connection

Select **Allow the connection** and click Next.

Step 5: Select Profiles

Choose the required Domain, Private, and Public profiles, then click Next.

Step 6: Name the Rule

Enter the name Allow Chrome Inbound TCP 443 and click Finish.

Step 7: Configure Port 443

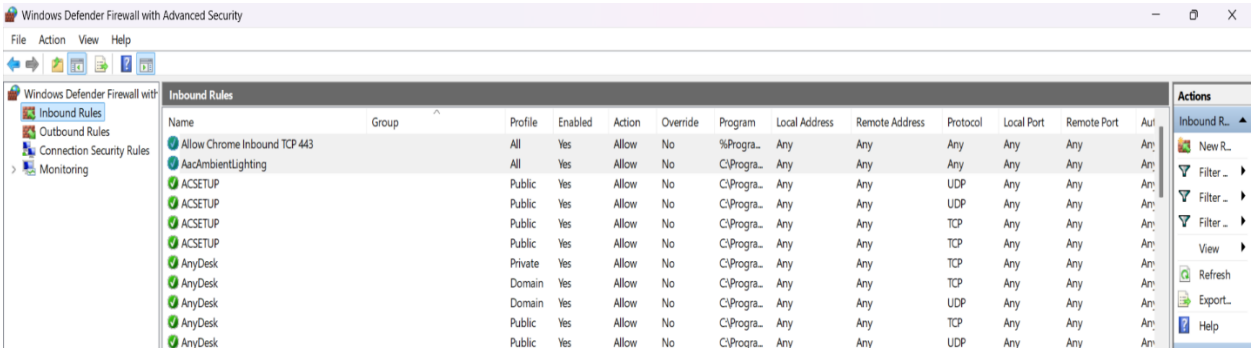
To meet both requirements, the rule must also restrict the allowed traffic to TCP port 443. Verify that the effective firewall configuration specifies both the Chrome program path and local TCP port 443.

Result

An inbound firewall rule was created for Google Chrome. The configuration must be verified to ensure that it allows only the intended inbound TCP traffic on port 443.

Conclusion

Windows Defender Firewall supports program-based and port-based rules to control incoming network traffic. Combining the application path with the required port provides more specific control than allowing all inbound traffic for an application.



Task 3 : Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

To block all incoming traffic from the IP address 123.45.67.89 using the Scope tab in Windows Defender Firewall.

Procedure

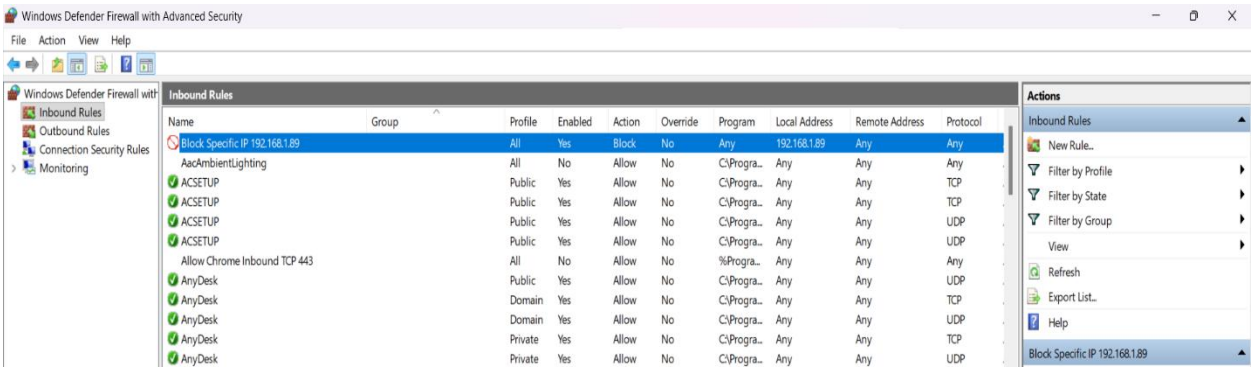
1. Open **Windows Defender Firewall with Advanced Security** by running wf.msc.
2. Select **Inbound Rules** and click **New Rule**.
3. Select **Custom** and choose **All programs**.
4. Set Protocol Type to **Any**.
5. In the **Scope** tab, select **These IP addresses** under remote IP addresses.
6. Click **Add** and enter 123.45.67.89.
7. Select **Block the connection** under Action.
8. Select Domain, Private, and Public profiles.
9. Enter the rule name Block Specific IP 123.45.67.89 and click **Finish**.

Result

A firewall rule was created to block incoming traffic from the specified IP address.

Conclusion

Windows Defender Firewall can block traffic from a specific remote IP address using the Scope tab, improving network security by restricting unwanted connections.



Task 4 : Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

To understand the differences between hardware and software firewalls and their use cases.

Comparison Table

Basis	Hardware Firewall	Software Firewall
Definition	A physical device that filters network traffic.	A program that filters network traffic on a computer.
Installation	Installed between a network and the internet.	Installed on an individual computer or server.
Protection	Protects multiple devices across a network.	Protects the device on which it is installed.
Cost	Usually more expensive.	Often included with the operating system or available at low cost.
Example	Dedicated firewall appliance or firewall-enabled router.	Windows Defender Firewall.

Use Cases

Hardware Firewall:

- Used in offices, schools, and large organizations.
- Protects multiple computers connected to a network.
- Helps control traffic entering and leaving the network.

Software Firewall:

- Used on personal computers and laptops.
- Blocks unauthorized incoming connections.
- Allows users to create rules for specific applications and ports.